

1 What should I submit, where should I submit and by when?

Your submission for this minor assignment will be one PDF (.pdf) file only. Instructions on how to prepare and submit this file are given below.

Assignment Package:

<https://www.cse.iitk.ac.in/users/purushot/courses/ml/2025-26-w/material/assignments/minor2.zip>

Deadline for all submissions: April 26, 2026, 9:59PM IST

Code Submission: *None needed – do not submit any code (ipynb, py, etc) files for this minor assignment*

Report Submission: on Gradescope

There is no provision for “late submission” for this minor assignment.

1.1 How to submit the PDF report file

1. The PDF file must be submitted using Gradescope in the *group submission mode*.
2. Note that unregistered auditors cannot make submissions to this minor assignment.
3. Make only one submission per assignment group on Gradescope, not one submission per student. Gradescope allows you to submit in groups - please use this feature to make a group submission.
4. Link all group members in your group submission. If you miss out on a group member while submitting, Gradescope will think that person never submitted anything.
5. You may overwrite your group’s submission as many times as you want before the deadline (submitting again on Gradescope simply overwrites the old submission).
6. Do not submit Microsoft Word or text files. Prepare your report in PDF using the style file we have provided (instructions on formatting given later).

Problem 2.1 (Melbo Mends Missing Images). Melbo is mesmerized by how even simple generative models such as single Gaussians offer reasonable prediction accuracies in the presence of missing pixels and also reconstruct the missing pixels. Melbo wishes to explore further and requests your help in doing so. For this task, you would need to refer to the following resources:

1. Calculations and code for the lecture “Generative Models” available at <https://www.cse.iitk.ac.in/users/purushot/courses/ml/2025-26-w/material/lectures/lec15.zip>
2. The MNIST dataset available at <https://www.kaggle.com/datasets/hojjatk/mnist-dataset/data>

(50 marks)

Your Task. The following enumerates 7 parts to the question. All parts need to be answered in the PDF file containing your report. **Do not submit any code files for this minor assignment.** Thus, submit only a single PDF file as your submission. The lecture code models each class as a single Gaussian i.e. $\mathbb{P}[\mathbf{x} | y = c] = \mathcal{N}(\boldsymbol{\mu}^c, \Sigma^c), c \in [C]$. Notice that the lecture code handles test images with missing pixels (where the pixels $\mathbf{x}^o$ are observed and the pixels $\mathbf{x}^m$ are missing) in a two-step manner:

- Step 1: make a prediction $\hat{y} = \arg \max_c \mathbb{P}[y = c | \mathbf{x}^o] = \arg \max_c \mathbb{P}[\mathbf{x}^o | y = c] \cdot \mathbb{P}[y = c]$
- Step 2: reconstruct using the prediction $\hat{\mathbf{x}}^m = \arg \max_{\mathbf{v}} \mathbb{P}[\mathbf{v} | \mathbf{x}^o, y = \hat{y}]$

Note that for all parts below, you need to use only a single Gaussian per class, not a mixture of Gaussians. Also, do not change the training procedure in the lecture code. Only change the testing/inference-related functions and the code for the experiment with missing pixels.

1. Melbo's friend Melba claims that one may get better results if one does a joint optimization i.e. solve $\{\hat{y}, \hat{\mathbf{x}}^m\} = \arg \max_{c, \mathbf{v}} \mathbb{P}[y = c, \mathbf{v} | \mathbf{x}^o]$. Give detailed calculations on how to perform this single-step inference efficiently. Are the single-step and two-step inference procedures identical? *Hint: nothing complicated like EM or alternating optimization is needed here – a simple application of product rule should suffice. You may find the following result useful: If the vector $\mathbf{x} = [\mathbf{x}^o, \mathbf{x}^m] \sim \mathcal{N}(\boldsymbol{\mu}, \Sigma)$, then $\mathbb{P}[\mathbf{x}^m | \mathbf{x}^o] = \mathcal{N}(\bar{\boldsymbol{\mu}}, \bar{\Sigma})$ where $\bar{\boldsymbol{\mu}} = \boldsymbol{\mu}^m + \Sigma_{mo} (\Sigma_{oo})^{-1} (\mathbf{x}^o - \boldsymbol{\mu}^o)$ and $\bar{\Sigma} = \Sigma_{mm} - \Sigma_{mo} (\Sigma_{oo})^{-1} \Sigma_{om}$.* (15 marks)
2. Describe what modifications need to be made to the code to implement your modified technique (e.g. the functions `predictClassScores` or `predictGen` may need to be merged into a single function). Give pseudo code or actual Python code in your PDF submission itself (do not submit code files separately). Be careful while choosing library functions to implement matrix inverses. (15 marks)
3. Compare the performance of your single-step prediction code with the two-step prediction code in the lecture with respect to classification performance, inference time and reconstruction quality. Repeat the experiment with various levels of censoring by modifying the function `sensorImages` to remove varying portions of the central portion of the image as shown below. Plot a graph showing how the lecture code and your code differ in classification performance and inference time as censoring becomes less or more aggressive. Describe in words how does reconstruction quality of the two methods vary as censoring becomes less or more aggressive. (20 marks)
 - `X[:, 4:24, 4:24] = 0` (51% missing pixels)
 - `X[:, 6:22, 6:22] = 0` (32% missing pixels)
 - `X[:, 8:20, 8:20] = 0` (18% missing pixels)
 - `X[:, 10:18, 10:18] = 0` (8% missing pixels)
 - `X[:, 12:16, 12:16] = 0` (2% missing pixels)

Using Internet Resources. You are allowed to refer to textbooks, internet sources, research papers to find out more about this problem and for specific derivations e.g. the arbiter-PUF problem. However, if you do use any such resource, cite it in your PDF file. There is no penalty for using external resources with attribution but claiming someone else's work (e.g. a book or a research paper) as one's own work without crediting the original author will attract penalties.

Restrictions on Code Usage. You should not use any library other than what is already imported in the Jupyter notebook in the assignment package (you do not need to anyway). Since this minor assignment does not require any code submission, we will trust you on this one. However, you may use fancier libraries like seaborn or plotly to plot graphs etc.

2 How to Prepare the PDF File

Use the following style file to prepare your report. The ZIP file contains an example file and instructions.

<https://media.neurips.cc/Conferences/NeurIPS2025/Styles.zip>

You must use the following command in the preamble

```
\usepackage[preprint]{neurips_2025}
```

instead of `\usepackage[preprint]{neurips_2025}`. Use proper $\text{\LaTeX}$ commands to neatly typeset your responses to the various parts of the problem. Use neat math expressions to typeset your derivations. Remember that all parts of the question need to be answered in the PDF file. All plots must be generated electronically - hand-drawn plots are unacceptable. All plots must have axes titles and a legend indicating what are the plotted quantities. Insert the plot into the PDF file using $\text{\LaTeX}$ `\includegraphics` commands.